Statistical Learning II

Lecture 6 - Bias-Variance decomposition (continued)

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Risk of OLS

Therefore, we have the following final result for the excess risk of OLS

$$\mathbb{E}_{\boldsymbol{\varepsilon}}\left[\mathscr{R}(\hat{\boldsymbol{\theta}}_{OLS})\right] - \sigma^2 = \sigma^2 \frac{d}{n}$$

Remarks:

- Excess risk is proportional to the noise level $\mathbb{E}[\varepsilon^2] = \sigma^2$.
- Excess risk is proportional to the data dimension.
- To achieve excess risk $\Delta\mathscr{R} < \delta$, need:

$$n > \frac{\sigma^2 d}{\delta}$$

samples.